

Bond polarity

Learning outcomes

By the end of this section you should be able to:

- Deduce the polar nature of a covalent bond from electronegativity values.
- Recognise bond dipoles represented with partial charges and vectors.

It's sports day in school and six students have decided to play against each other in a game of tug-of-war (**Figure 1**). They split into two equal teams of three students, and begin. No matter how hard either team pulls, they cannot get the ribbon over the line. Why do you think this is? How could this be explained in terms of forces?

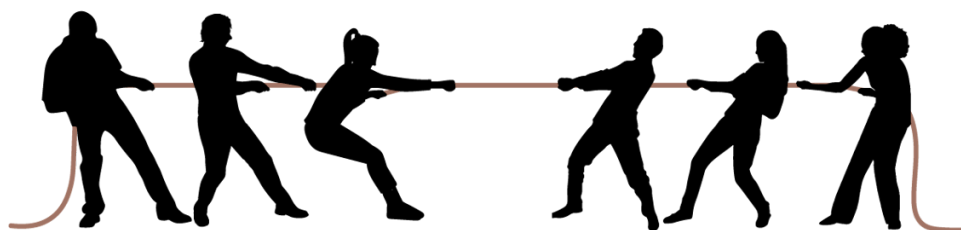


Figure 1. Evenly balanced teams exerting the same pull on the rope in tug-of-war.

Two students grow tired of the game and decide to sit down. Within the seconds the game had ended, the difference in pull that each team could exert was significantly different and the team with the stronger pull won (**Figure 2**).

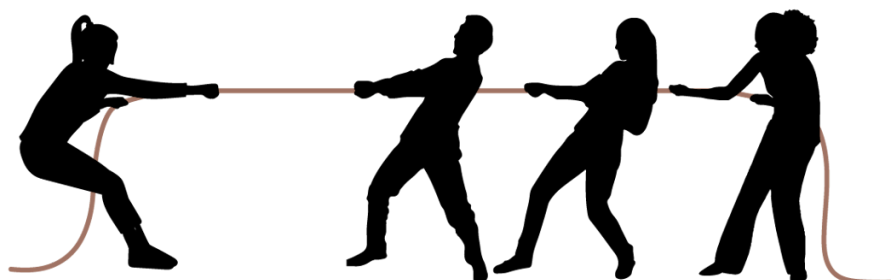


Figure 2. Uneven teams exerting different amount of pull on the rope in tug-of-war.

 More information for figure 2

The illustration depicts a tug-of-war between two teams of unequal numbers, pulling on opposite ends of a rope. One team has more participants than the other, showing an imbalance in strength. This image is used as an analogy to explain covalent bonds, where electrons are shared unevenly between two nuclei due to differing attraction strengths, similar to the uneven pull in the tug-of-war. The image conveys the concept of electrostatic attraction and shared electrons in covalent bonds.

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Think about what you know about covalent bonds: electrons are being shared between two nuclei and there is an electrostatic attraction between the nuclei in each atom and the shared electrons. Do you think they are always held at an equal distance between the two nuclei? Is it possible for one nucleus to have a stronger attraction than the other?

Electronegativity

In section S2.2.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>), you learned that a covalent bond is the electrostatic attraction between the nuclei of two atoms and a shared pair of electrons. What creates the electrostatic attraction?

The opposite charges of the positively charged nuclei and negatively charged electrons cause this electrostatic attraction.

Let's visualise this using a simple molecule of hydrogen, H₂ (**Figure 3**).

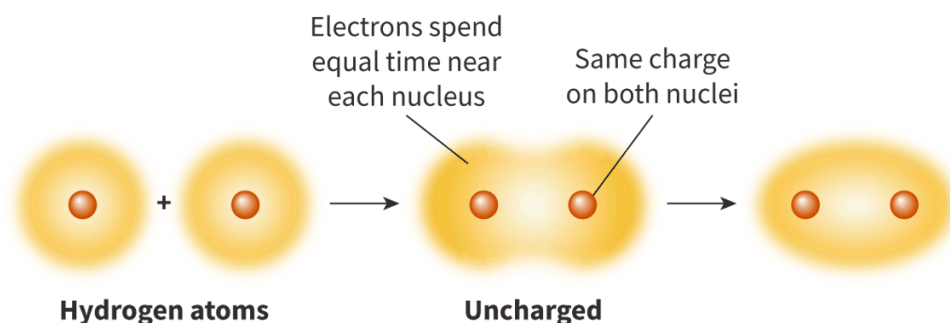


Figure 3. Electrons are held at equal distances between the two hydrogen nuclei and so the electron cloud demonstrates this even distribution.

 More information for figure 3

The image is a diagram illustrating the distribution of electrons in a hydrogen molecule. It consists of three stages:

1. **Left Side:** Two separate yellow spheres representing hydrogen atoms with visible small brown nuclei and a plus sign between them to indicate separation. Below, the text states "Hydrogen atoms."
2. **Middle:** The two spheres are now closer, overlapping slightly, forming a larger combined shape. Lines indicate the text: "Electrons spend equal time near each nucleus" and "Uncharged" below the overlap.
3. **Right Side:** A single larger, elongated yellow shape, symbolizing the hydrogen molecule with equally distributed electrons. It has annotations pointing to both nuclei with the text: "Same charge on both nuclei."

This diagram visualizes the equal distribution of electrons between the two hydrogen nuclei in a molecule, illustrating the concept of a covalent bond where electrons are shared equally, resulting in no net charge difference.

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There is one pair of electrons being shared between two atoms of hydrogen. Both hydrogen atoms have one positive proton in their nuclei. Where do you think the shared electrons are likely to be found?

Although electrons are continuously moving in random directions, they are most likely to be found at an equal distance between those two nuclei.

Let's compare this to a different molecule, hydrochloric acid or HCl, which is shown in **Figure 4**. The hydrogen atom is on the left and chlorine on the right. Which side appears to have a greater electron cloud?

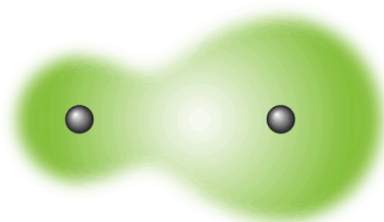


Figure 4. The electron cloud for HCl demonstrates an unequal electron distribution.

 More information for figure 4

The image shows the electron cloud distribution in a hydrochloric acid (HCl) molecule. The illustration depicts two circles connected in the center, representing the hydrogen atom on the left and the chlorine atom on the right. The circles depict the regions of electron density, with a more significant electron cloud around the chlorine atom, indicating unequal electron distribution. The electromagnetic cloud is broader and denser around the chlorine compared to the hydrogen, visually demonstrating the molecule's polarity.

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The electron cloud surrounding the side of the molecule with chlorine is greater. This indicates that the electrons are being found closer (on average) to the chlorine nucleus.

Why do you think this is? What is different about these two atoms?

Chlorine has more protons in its nucleus than hydrogen does, so it can exert a greater electrostatic attraction, but this is not the only factor that affects electrostatic attraction.

Chlorine also has a larger atomic radius as it has more energy levels occupied by electrons, and these other electrons also offset some of the effect of the nucleus. You will learn more about both of these concepts in [section S3.1.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/>).

There are many factors that affect an atom's ability to electrostatically attract a shared pair of electrons. So, a new scale of measurement is needed: electronegativity.

Study skills

You can find a periodic table containing the values for electronegativity in [section 9](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet.

Electronegativity is defined as an atom's ability to attract a covalently bonded pair of electrons. It is measured on a relative scale that ranks elements on the basis of this ability, with fluorine being the most electronegative element and francium being the least. You will learn more about the trend that exists for

electronegativity when you study [section S3.1.3](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/>).

The most commonly used scale, the Pauling scale, was developed by Linus Pauling in 1932 (**Figure 5**). It ranges from 0 to 4 and although the values are typically quoted as absolute numbers, if units are assigned then they are ‘Pauling units’.

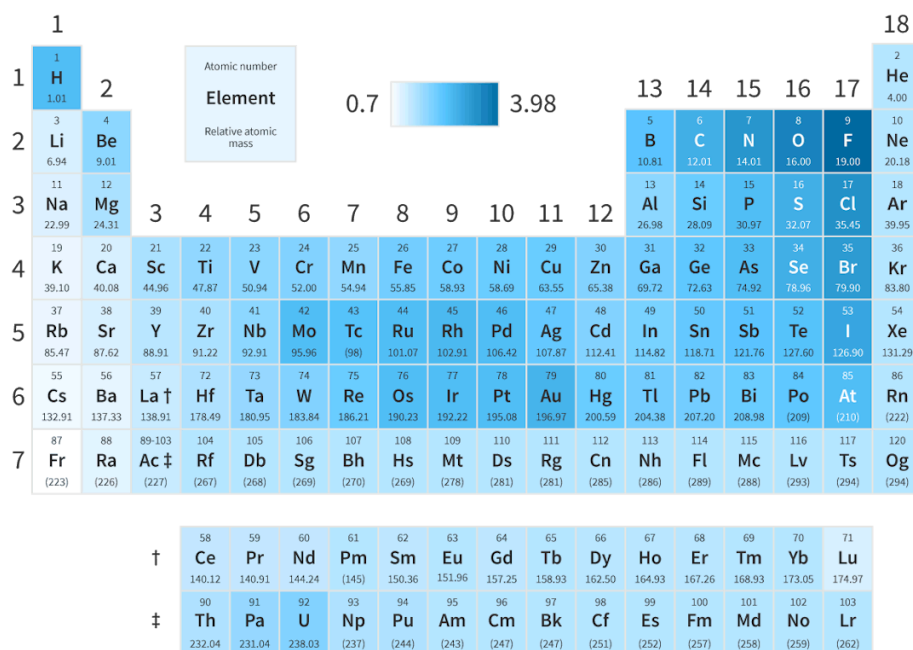


Figure 5. Pauling electronegativity scale.

More information for figure 5

The image is a periodic table that color codes elements according to Pauling electronegativity values. Each cell represents an element and contains its symbol, atomic number, and relative atomic mass. A gradient color bar at the top indicates the electronegativity scale, ranging from 0.7 to 3.98, with different shades of blue representing varying electronegativity values. Elements like Fluorine have a higher electronegativity (3.98), while Caesium is lower (around 0.7).

The standard layout is divided into groups and periods. Groups are vertical columns numbered 1 to 18, while periods are horizontal rows numbered 1 to 7. Lanthanides and actinides are shown below the main table.

Important sections include: - Group 1: alkali metals, like Lithium (Li) with a low electronegativity. - Group 17: halogens, like Fluorine (F) with the highest electronegativity. - Noble gases (Group 18) typically have undefined electronegativity in the Pauling scale.

The table efficiently represents element trends in properties and behaviors based on their electronegativity differences.

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Exercise 1

Click a question to answer



1. Use the data in [section 9](#) (<https://app.kognity.com/:sectionlink:137929>) of the DP chemistry data booklet to complete the statement: the most electronegative element is _____ which has a value of _____.

☒ Check answers

By understanding electronegativity and the Pauling scale, you can start to see why some elements bond how they do, and how an asymmetrical distribution of electrons within a bond could affect the property of that bond.

Polar and non-polar bonds

When you are considering the bond between two atoms, it is important to remember that each atom has its own electronegativity value. It is the **difference** in electronegativity that causes asymmetrical electron distribution.

For example, the hydrogen molecule, H_2 , is composed of two hydrogen atoms with the same electronegativity values and so the pair of electrons is equally attracted to both hydrogens. O_2 is another molecule composed of two atoms with the same electronegativity values, with the pairs of electrons equally attracted to both oxygens (**Figure 6**). Think of other examples that would have an electronegativity difference of 0.

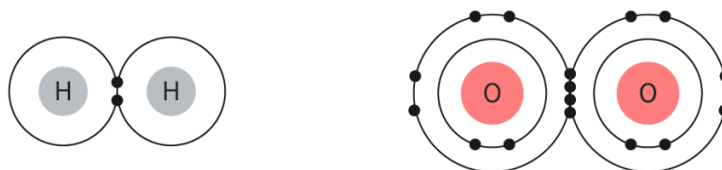


Figure 6. Hydrogen and oxygen molecules have non-polar bonds.

 More information for figure 6

The image is a diagram depicting non-polar bonds in hydrogen and oxygen molecules. On the left, there are two hydrogen atoms, each labeled with an 'H', and they are joined together with a non-polar bond, indicated by overlapping circles that represent shared electrons equally distributed between the atoms. On the right, two oxygen atoms are shown, each labeled with an 'O', also joined by a non-polar bond. Like the hydrogen atoms, the oxygen atoms share electrons equally, depicted by the visual arrangement of electron orbitals around each atom. The diagram illustrates the concept of non-polar bonding due to equal electronegativity in both molecular structures.

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We call these types of bond non-polar bonds. They are composed of atoms with the same or similar electronegativity values, and have similar levels of attraction for their shared pair(s) of electrons. The electrons in a non-polar bond are located at equal distances between the two atoms sharing them, on average.

The electronegativity values of the atoms sharing electrons does not have to be exactly the same, but there should not be a large difference between them. An electronegativity difference of 0.1 to 0.4 results in a bond that is in the range of non-polar to weakly polar, as the difference increases. As a guide, the difference in values is typically no more than 0.4 in a non-polar bond (see **Table 3**). However, it should be noted that even a bond with a difference of 0.1 electronegativity units will be ever so slightly polar and the polarity of the bond increases as the difference in electronegativity increases.

Table 3. Polar vs. non-polar bonds.

Polar bonds	Non-polar bonds
Between atoms with different electronegativity values (typically a difference of 0.4—1.7)	Between atoms with the same or similar electronegativity values
Bonded electrons found closer to one atom than the other atom	Bonded electrons found equal distance between two atoms sharing them

A polar covalent bond is formed when there is a reasonably significant difference in electronegativity values of the two atoms being bonded. This means that one of the atoms has a much greater attraction for the shared electrons than the other atom so they are found closer to one atom than the other. Because of this, the atom with stronger attraction will have greater electron density than the atom with a lower attraction (**Figure 7**).

Take a water molecule for example, H_2O . Hydrogen has an electronegativity value of 2.2 and oxygen has an electronegativity of 3.4. This indicates that oxygen can exert a much stronger attraction on a shared pair of electrons than hydrogen can. When these atoms are bonded together the pair of electrons being shared between hydrogen and oxygen will be found closer to the oxygen molecule, and the oxygen will have greater electron density than hydrogen.

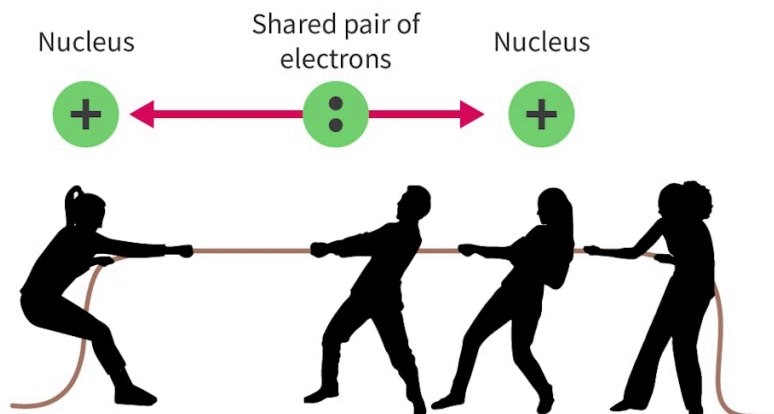


Figure 7. Polar covalent bond.

 More information for figure 7

The image is a diagram illustrating the concept of a polar covalent bond using a tug-of-war analogy. On the left side, there is a person pulling on a rope, representing a nucleus with a positive charge. In the center, there is an illustration of a "shared pair of electrons" indicated by a dot pair, with arrows pointing towards each nucleus, depicting the attraction. On the right side, three people are pulling back on the rope, symbolizing another nucleus. The diagram uses visuals to show the imbalance in electron sharing, where one side is pulling harder, similar to a tug-of-war game, indicating higher electronegativity and electron density on one side of the bond.

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The shift in location of these electrons can affect the overall properties of that bond as one side of the bond is now 'electron-rich' while the other side is 'electron-poor'. As there have been no ions formed, using a + or – to indicate this difference in electron density would be inappropriate. Instead, to indicate the partial charge formed the symbol delta, δ , is used. In this context, δ means partially or slightly (**Figure 8**).

The atom with greater electron density due to its higher electronegativity is considered to be δ^- while the atom with lower electronegativity is considered to be δ^+ .

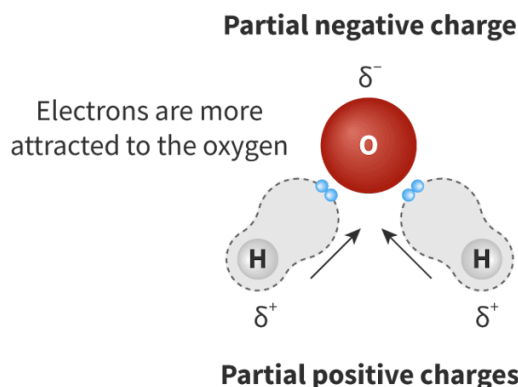


Figure 8. Polar water molecule resulting from a difference in electronegativity values between hydrogen and oxygen.

More information for figure 8

The image is a diagram depicting the polar nature of a water molecule. At the center is an oxygen atom labeled 'O', which has a partial negative charge (δ^-) due to greater electron density. Two hydrogen atoms, each labeled 'H', are symmetrically positioned on either side of the oxygen atom, each with a partial positive charge (δ^+). Arrows point towards the oxygen atom, indicating that electrons are more attracted to the oxygen. The diagram includes text that states 'Partial negative charge' above the oxygen and 'Partial positive charges' below the hydrogen atoms. The phrase 'Electrons are more attracted to the oxygen' is also shown next to the arrows pointing to the oxygen atom, illustrating the molecule's polarity.

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Typically, a difference in electronegativity values of between 0.4 and 1.7 will result in a polar covalent bond. At this point, there is a significant enough difference in their relative attraction for the shared pair of electrons. What do you think would happen if the difference in electronegativities was greater than 1.7?

Making connections

As part of [section S2.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/), you learned that ionic bonds form when there is a complete electron transfer, one atom gains electrons while the other atom loses them. The atoms were able to do this as they had a large difference in their electronegativity values, the cation had a much lower electronegativity value than the anion did. The difference in their attraction for the electrons is what resulted in one of them gaining full control of the electrons while the other one lost control.

This is actually the determining factor for if an ionic or covalent bond is formed between two atoms; it's all due to the electronegativity difference between the two species being bonded. If the difference is greater than 1.7, then the bond formed will be ionic (**Figure 9**).

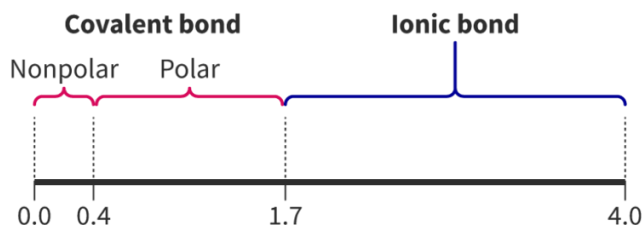


Figure 9. Scale of electronegativity difference to determine whether ionic or covalent bond is formed.

 More information for figure 9

The image shows a scale representing the differences in electronegativity used to determine the type of bond formed between atoms. The scale ranges from 0.0 to 4.0.

- From 0.0 to 0.4: Nonpolar covalent bonds, indicated by a pink bracket.
- From 0.4 to 1.7: Polar covalent bonds, also shown with a pink bracket.
- From 1.7 to 4.0: Ionic bonds, highlighted with a blue bracket.

The scale helps determine that if the electronegativity difference is less than 1.7, a covalent bond forms (nonpolar if under 0.4 and polar if between 0.4 and 1.7). If the difference is greater than 1.7, an ionic bond forms.

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The partial charges associated with polar bonds are often referred to as a dipole or a dipole moment. In **Figure 9**, oxygen has a partial negative charge and hydrogen has a partial positive charge. There are a number of different ways in which a dipole can be represented (**Figure 10**).

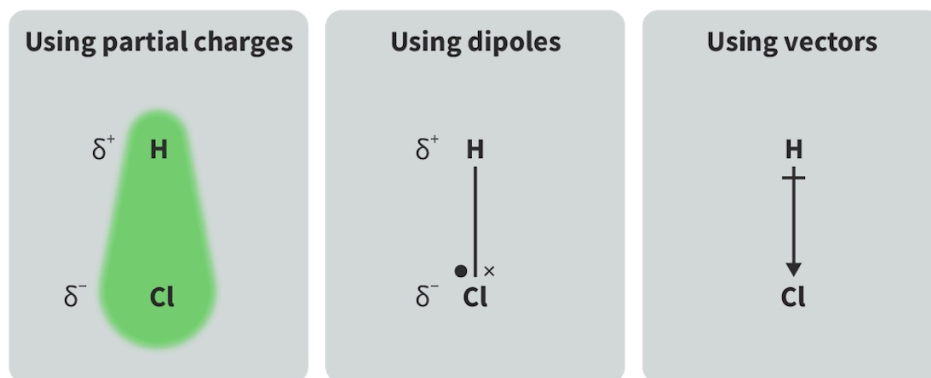


Figure 10. Methods of representing polar bonds.

 More information for figure 10

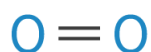
The image shows three panels illustrating different methods of representing polar bonds. The first panel, labeled 'Using partial charges,' depicts two atoms: hydrogen (H) with a partial positive charge (δ^+) and chlorine (Cl) with a partial negative charge (δ^-), connected with a shaded region indicating the partial charge difference.

The second panel is titled 'Using dipoles' and shows the same two atoms, H with a δ^+ and Cl with a δ^- , connected by a straight line. An arrow crosses the line near Cl, indicating the direction of the dipole moment from H to Cl.

The third panel, labeled 'Using vectors,' illustrates H and Cl with an arrow pointing from H to Cl, representing the vector direction of the polar bond.

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Use section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) in the DP chemistry data booklet to help you complete this activity in **Interactive 1**. You should drag and drop a partial charge on to each atom in the bonds below. If the bond formed is not polar, drag the 'No δ ' label to each atom in that bond.

 δ^+ δ^- No δ

✓ Check

Interactive 1. Dipole or No Dipole?

In this next activity, you will create 'profile cards' for different elements practicing how to determine the type of bond formed between different pairs of elements.

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:**
 - Social skills — Working collaboratively to achieve a common goal
 - Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Group activity

Profile cards

Materials:

- Element cards printed [Element cards](https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemical%20Elements%20Cards.pdf) 
(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemical%20Elements%20Cards.pdf)
- String 2 m long (per pair of students)
- Styrofoam balls × 2 (per pair of students)

Procedure:

1. Before you begin this activity, you should spend 5 minutes (maximum) completing the 'profile' card for the element(s) you have been assigned. Use [section 9](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) in the DP chemistry data booklet to help you complete this.
2. Now with a partner, calculate the electronegativity difference between your elements and identify the type of bond formed.
3. Collect the bond information from all student pairs in a table on the whiteboard, similar to the one shown here.

Element 1	Element 2	Electronegativity difference
Nitrogen	Oxygen	
Nitrogen	Chlorine	
Phosphorous	Chlorine	
Fluorine	Fluorine	
Carbon	Oxygen	

Element 1	Element 2	Electronegativity difference
etc.		
etc.		

1. Now pair up with a different student and repeat steps 2 and 3. Continue to do this until all possible bond combinations have been reached.

Representing these bonds:

You can recreate these bond pairing using Styrofoam balls to represent electrons and a string to hold the electrons in the correct arrangement.

The side of the bond that has a lower electronegativity should hold the string up higher as gravity will force the two electrons (Styrofoam balls) to move closer to the higher electronegativity side (see **Figure 11**), representing the polarity of the bond.

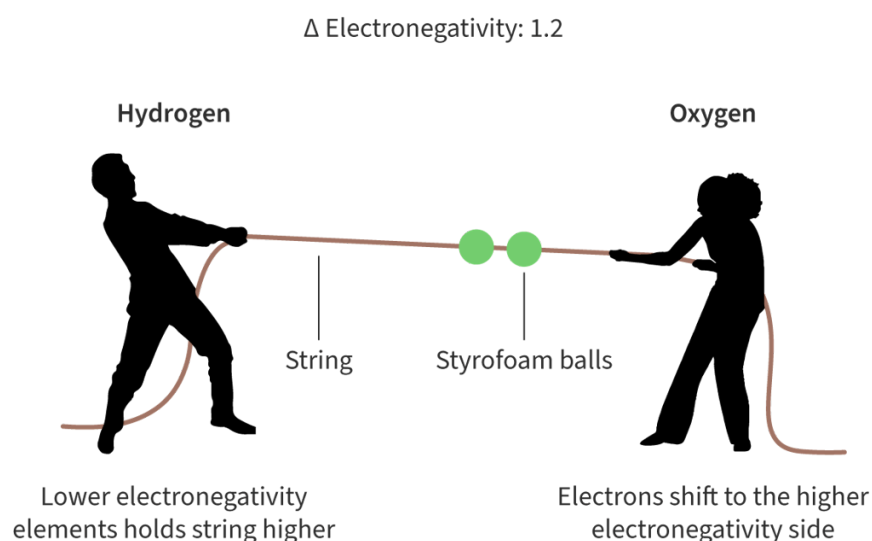


Figure 11. Two people demonstrate asymmetrical electron distribution based in electronegativity values.

 More information for figure 11

The diagram illustrates the concept of asymmetrical electron distribution in a polar bond using Styrofoam balls and a string. On the left side, the element labeled 'Hydrogen' has a lower electronegativity. This side holds the string higher, indicating that electrons are more attracted to elements with higher electronegativity. Moving right, the two green Styrofoam balls represent shared electrons in a bond. The string, representing bond alignment, slopes downward toward the right side, which is labeled 'Oxygen.' The text ' Δ Electronegativity: 1.2' is placed above, indicating the difference in electronegativity between the two elements. The text underneath notes that 'Lower electronegativity elements hold string higher,' and 'Electrons shift to the higher electronegativity side,' visually suggesting electron shift towards Oxygen due to its higher electronegativity.

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